

WILLIAM DUAN

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SUMMARY

Economist finishing a PhD at UBC in 2027. I specialize in auctions and mechanism design. My thesis studies markets such as online ad auctions, where bidders cannot see the allocation rule and learn only from whether they win. Five years teaching PhD-level microeconomic theory, with three departmental teaching awards.

EDUCATION

PhD in Economics, University of British Columbia 2020 – 2027 (expected)
Fields: microeconomic theory, information economics, game theory. SSHRC Doctoral Fellowship, 2020 – 2024.

MA in Economics, University of British Columbia 2019 – 2020

BA, First Class Honours in Economics, McGill University 2015 – 2019
Minor in Mathematics and Computer Science. GPA 3.92.

RESEARCH

Selling Mechanisms under Misspecified Learning (job market paper) 2024 – present

- Models repeated auctions in which bidders do not observe the allocation rule, as with the Google ad auctions examined in the 2023 DOJ antitrust trial and the Amazon Buy Box.
- Shows when a seller can design the mechanism so that bidders' beliefs settle on a chosen target as they learn, and what this does to revenue relative to the standard optimal auction.
- Developed a computational method that checks the conditions and constructs such mechanisms numerically.
- Presented at the Canadian Economics Association Annual Meetings, 2026.

Ex-ante Heterogeneity in Directed Search Games (working paper) 2022

- Shows that the standard equilibrium of a job-application model breaks under small differences between workers, and gives a recursive algorithm that constructs a robust alternative.

Research Assistant, McGill University (Prof. Hassan Benchenkroun) 2017 – 2019

- Derived solutions to dynamic programming models and ran numerical simulations of trade and resource-extraction models.

TEACHING

Instructor and Course Designer, UBC Economics Math Camp Summer 2026

- Designed and taught the math preparation course for incoming MA and PhD students. Wrote the slides and problem sets in Quarto and kept them in a shared Git repository.

Teaching Assistant, Vancouver School of Economics 2019 – 2026

- Led tutorials and office hours for the PhD microeconomic theory sequence for five years, plus four undergraduate and graduate courses. Helped write the ECON 601 problem sets.
- Agustín Beltrán Award for Excellence in Teaching, 2024, 2025 and 2026.

SKILLS

Programming: Python, R, Julia, MATLAB, Mathematica, Java; LaTeX, Quarto, Git

Methods: auction and mechanism design, game theory, Bayesian learning, dynamic programming, numerical simulation

Languages: English and Mandarin (fluent), French (basic)

Citizenship: Canadian